New Datasets and Methods for `day-in-the-life` Continual Learning Benchmark

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Abstract

1 Introduction and Motivation: MARIA

Continual learning explores how AI systems can adapt and continue training and learning at deployment time. This property is necessary for systems to adapt to real-world scenarios with data distribution shifts and constantly changing information. Models operating in continual learning scenarios inherently struggle with (catastrophic) forgetting, and while there are multiple approaches for lessening or mitigating this phenomenon **?**, researchers in the area have not developed a comprehensive, real-world benchmark for continual learning tasks. Most papers we are familiar with in the field operate on MNIST or CIFAR datasets and are able to prove the effectiveness of the respective methods in this very particular setup. Our goal is to test continual learning methods on valuable and interesting tasks that resemble a day in a life of a human (hence the title): finding your way around in an environment, learning new skills, reasoning through hard questions, etc. This manuscript describes our contributions along three dimensions (infrastructure, datasets, and methods) to the larger `day-in-the-life` benchmark. Our main additions and findings are the following:

- Sth about DER and similarity stuff
- Sth about DualPrompt
- Sth about new datasets and infrastructure for them

2 Literature Review

Continual learning methods are usually classified into five main areas – they can be based on regularization, replay (also called rehearsal), optimization, representation, and architecture **?**. This classification is not strict, and there is often overlap between the categories. To give some examples of methods from different categories, EWC **?** falls under regularization, GEM **?** is a dual replay- and optimization-based approach, Side-Tuning **?** is a representational method, and Piggyback **?** is an architectural method. The methods we explore and implement for `day-in-the-life` fall under replay (DER **?**) and representation (DualPrompt **?**).

2.1 Experience Replay: CHRISTIAN

2.2 Similarity Weighting: CHRISTIAN

2.3 Prompting in Continual Learning: MARIA

Prompting appeared as a representational approach instructing a pretrained model to adapt to new knowledge in the continual learning setting in L2P **?**. Instead of relying on example memorization

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and replay, the knowledge about new tasks is injected into the model via training a small pool of parameters (called prompts) that are selected and added to input tokens.

DualPrompt **?** is a method inspired by L2P. It also adds a small set of new parameters (prompts) to train on top of a frozen backbone to instruct the large model to learn sequentially arriving tasks. Typically, prompts constitute 0.2-0.6% of the full model size. In DualPrompt, prompts are explicitly divided into task-specific and task-invariant instructions, which gives a sizable increase in performance as compared to the original L2P framework. All the parameters are trained together with a learnable classifier (the target task in the DualPrompt paper is classification). At test time, the prompt parameters are prepended to the embedding feature extracted from the frozen backbone model in the multi-head attention layers, which is then sent to the rest of the model.

3 Methodology

We are contributing to the `day-in-the-life` benchmark in three directions: methods that we test on the benchmark, the infrastructure of the benchmark's repository, and datasets. The new methods were tested on the `day-in-the-life`

3.1 Backbone Model

We considered multiple base models to benchmark, fine-tune and test our CL algorithms on; these included the Qwen, SmolVLM, and Gemma model families. During our discussions, we landed on a few highest priority characteristics for selecting the default base model: ease of use and access (open-source), size (to increase the accessibility of the benchmark), quality and breadth of documentation (for dependable baseline performance), and community adoption (familiarity, edge-case debugging, guidance). Taking these criteria into account, we arrived at Qwen3-VL-2B-Instruct. This model is well documented and popular within the community, supports vision and text, and small enough to fit on the commonly-accessible Nvidia L4. In addition, its performance on many datasets lies in the "sweet spot": not so strong that the model is saturated, obfuscating the results between different CL algorithms, and not so weak that additional training would yield negligible benefit. Finally, the suite of Qwen3-VL models range in size from 2B to 32B parameters, allowing benchmarkers with greater compute resources to upscale.

3.2 DER Implementation: CHRISTIAN

3.3 Similarity Weighting Implementation: CHRISTIAN

3.4 DualPrompt Implementation: MARIA

3.5 New Datasets: AKSEL

Due to the variety of datasets, GPUs, and contributors, a standard contribution and benchmarking pipeline had to be formed and agreed to. The per-dataset deliverable consisted of

1. A function to download and save the raw data
2. A pre-processing function to convert the data into desired form (i.e. raw tensor -> image)
3. A loader function to download the processed dataset from a predetermined location on disk
4. A configuration dictionary with context, prompting, and post-processing to ensure the LLM's answer format matched the dataset's provided answer format
5. Documentation on the performance of the benchmark LLM out-of-the-box, after 1 epoch of LoRA finetuning, and after 5 epochs of finetuning

3.6 Benchmarking Protocol and Evaluation Metrics: AKSEL

3.7 Changes to Processing Existing Datasets: Maria

4 Experimental Results

4.1 Qwen-4b Performance Reproduction: AKSEL

4.2 New Methods' Performance: CHIRSTIAN and MARIA

4.3 Ablations for Replay and Similarity Weighting: CHIRSTIAN

4.4 Learning Rate Ablations for Baseline Training: MARIA

4.5 DualPrompt Ablations: MARIA

5 Further Discussion

5.1 Challenges and Limitations: CHRISTIAN

5.2 Future Directions: AKSEL

6 Conclusion

6.1 Summary of Findings: ALL

6.2 Contributions: ALL

Contributions of each team member:

- Aksel:
- Christian:
- Maria:

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Code Availability

All the code used in this project is available at `https://github.com/AmoghSInamdar/day-in-the-life`.

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Appendix